OENG1209: Engineering Science

Week 11:

Oscillations & Waves

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Today's Topics

- Simple Harmonic Motion
- Modeling Simple Harmonic Motion with Circular motion
- Energy in Simple Harmonic Motion.
- The Travelling Wave & Sinusoidal Waves.
- Power, Intensity and Decibels.

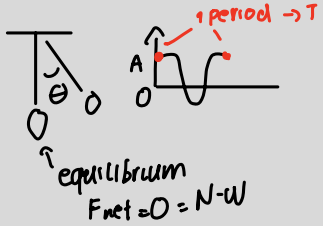
Simple Harmonic Motion

Amplitude and Frequency

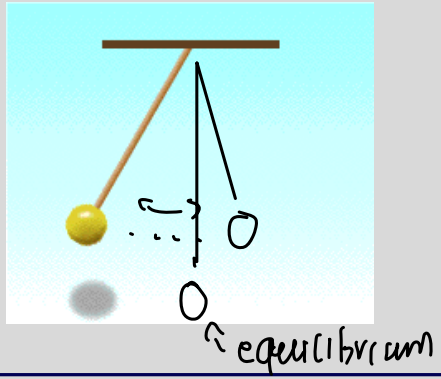
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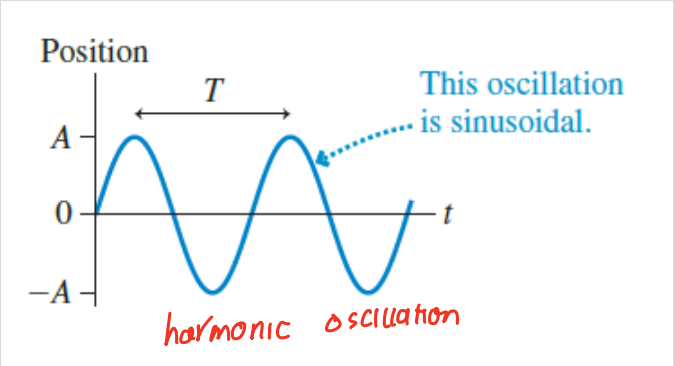
Oscillatory Motion

- **Oscillatory motion** is a type of **repetitive motion** back and forth around **an equilibrium position**.



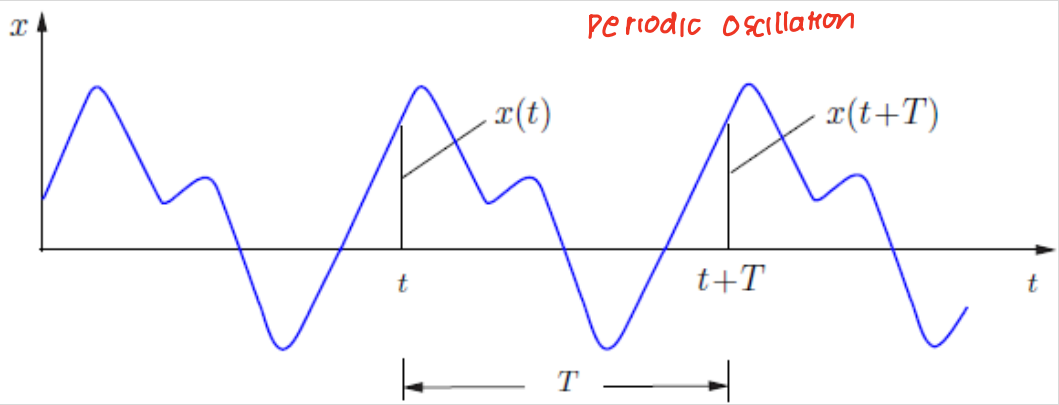
- All oscillators have two things in common:
  - The oscillation take place around an **equilibrium position**.
  - The motion is **periodic**, **repeating** at regular intervals of time.





Oscillatory Motion

Periodic vibration



The position $x(t)$ repeats itself after a time interval T:

$$x(t+T) = x(t)$$

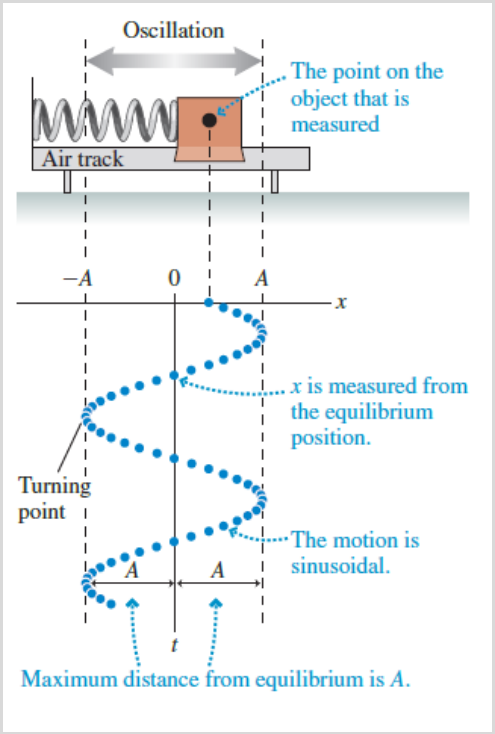
→ **PERIODIC** vibration

→ T is called the **PERIOD** of the vibration

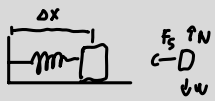
Oscillatory Motion

Simple Harmonic Motion

Consider the **Mass Spring system**.



Pos : $x(t) = ?$



$-F_s = F_{net} = ma$

$-kx = ma$

$-kx(t) = ma(t) = m\frac{d^2x(t)}{dt^2}$

$x = x(t)$ differencial

$-kx = m\frac{d^2x}{dt^2}$

$m\frac{d^2x}{dt^2} + kx = 0$

$\frac{d^2x}{dt^2} + \left(\frac{k}{m}\right)x = 0$, ω^2

$\Rightarrow x(t) = A\cos(\omega t)$ or $x(t) = B\sin(\omega t)$

$\frac{dx(t)}{dt} = -A\omega\sin(\omega t)$

$\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t)$

x → v $\frac{dx}{dt}$ → a $\frac{d}{dt}\frac{dx}{dt}$

$t = 0 \Rightarrow \begin{cases} x(0) = 0 \\ v(0) = C \text{ not } 0 \end{cases}$

$x(t) = B\sin(\omega t)$, $t=0 \to 1$

$v(t) = \frac{dx}{dt} = B\omega\cos(\omega t) = C$

$\Rightarrow B = C/\omega$

Init condition

$t = 0 \Rightarrow x(0) = A$

$v(0) = 0$

$\Rightarrow x(t) = A\cos(\omega t)$

$x(0) = A(1) = A$ ✓

$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = -A\omega^2\cos(\omega t) + \omega^2 A\cos(\omega t) = 0$

Simple Harmonic Motion

- The **period** (T) is the time it takes to **complete one full cycle** of oscillation.
- The **Amplitude** (A) is the maximum displacement from equilibrium

$\omega^2 = k/m$

$x(t) = B\sin(\omega t)$

Find: relation T & ω

Condi: $x(t+T) = x(t)$

$\begin{cases} y(t) = B\sin(\omega t) \\ x(t+T) = B\sin(\omega(t+T)) \end{cases}$

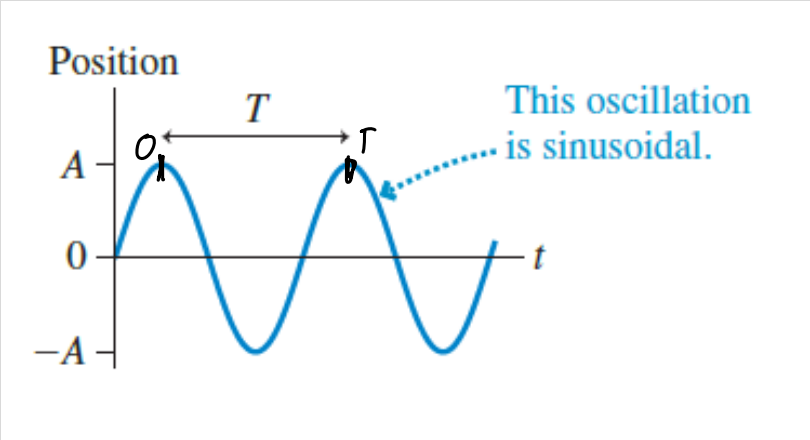


$\omega(t+T)$ $\quad 2\pi$ ↑ 1rev

$\omega(t+T) = \omega t + 1\text{rev}$

$\omega T = 2\pi$

or $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$



Simple Harmonic Motion

- The **frequency (f)** is the number of cycles or oscillations completed per second.

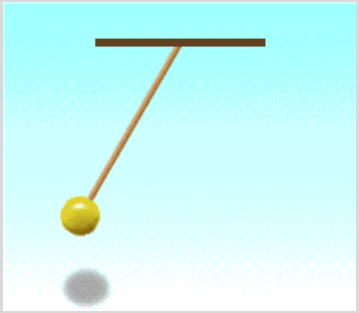
T: time for 1 cycle 1 cycle per T $\frac{1}{T}$ cycle per time

- **The frequency** is the inverse of the period.

$$f = \frac{1}{T}$$

- The unit of frequency is **Hertz (Hz).**

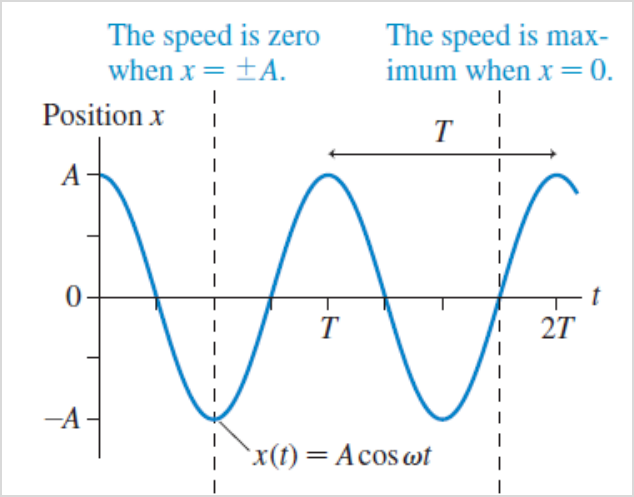
$$\mathbf{1\ Hz = 1\ cycler\ per\ second = 1\ s^{-1}}$$

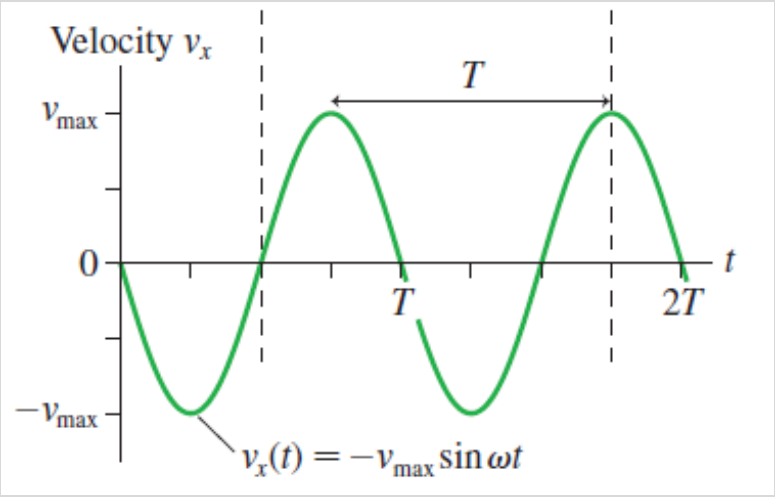


Simple Harmonic Motion

$x(0) = A$, we can deduce that the motion will be defined by a cosine, specifically:

$$x(t) = A\cos\omega t$$





Simple Harmonic Motion – Example

Example: A block is attached to a horizontal spring resting on a frictionless surface. Someone pulled the block 20.0 cm to the right, and released it at $t = 0s$. It makes 15 oscillations in 10.0 seconds.

a. What is the period of oscillation for the system?

b. What is the block's maximum speed? "no mag

c. What are the position and velocity of the block at $t = 0.800\ s$?

a) 20cm 15 cycles → 10s
1 cycle → Ts

$T = \frac{10}{15} = 0.667s$

b) $x = A\cos(\omega t)$

$v = \frac{dx}{dt} \left[-A\omega\right]\sin(\omega t)$

$v_{max} = 20cm \cdot \frac{2\pi}{0.667}$ $\omega = 2\pi/T$

c) $t = 0.8s$

$\Rightarrow x = 0.2\cos\left(\frac{2\pi}{0.667} \cdot 0.8s\right)$

↓

$v =$

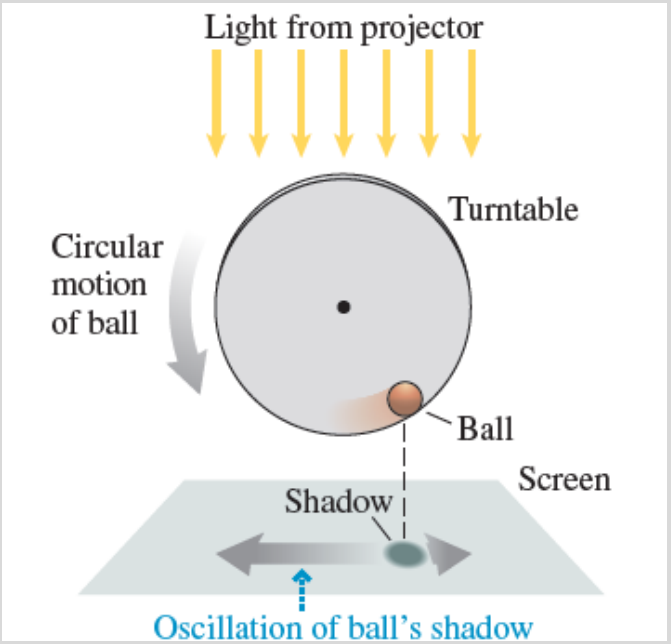
Simple Harmonic Motion and Circular Motion

Using circular motion to model harmonic motion.

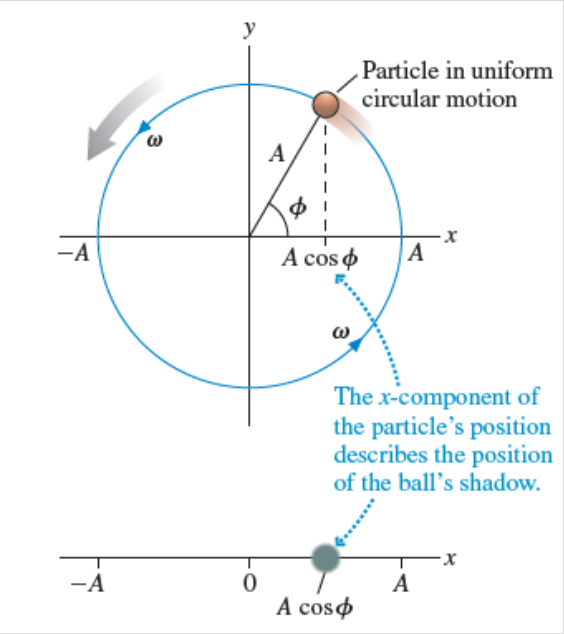
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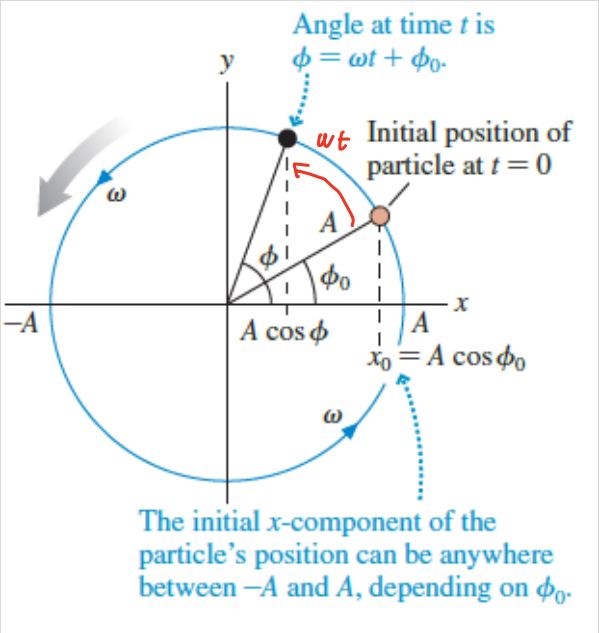
SHM and Circular Motion

Consider **the motion of a ball** running around a transparent turntable. A projector is shining a strong light on the table such that **the shadow of the ball** is casted onto a screen.



SHM and Circular Motion





$$x(t) = A\cos(\omega t + \phi_0)$$

$\phi = \phi_0 + \omega t$

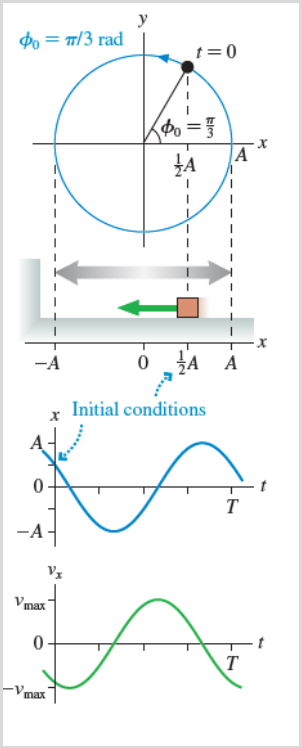
$x = A\cos(\phi_0 + \omega t)$

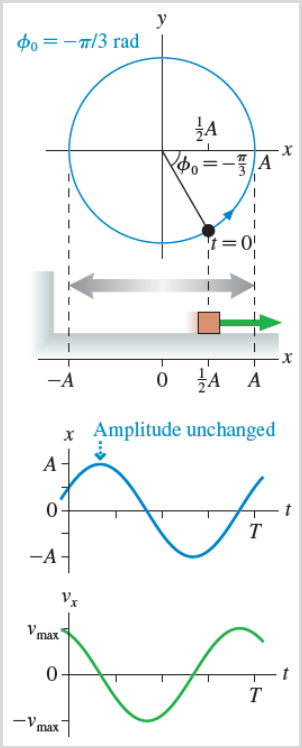
$v = -A\omega \sin(\phi_0 + \omega t)$

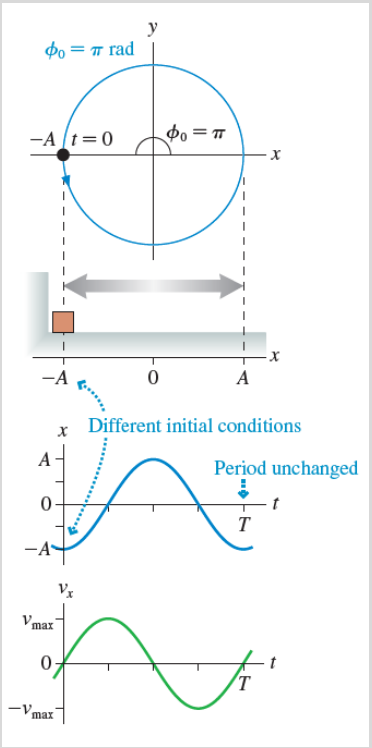
we can say $\omega = 2\pi f$ (f: angular frequency)

SHM and Circular Motion

Examples of different initial conditions are described by different values of the phase constant:







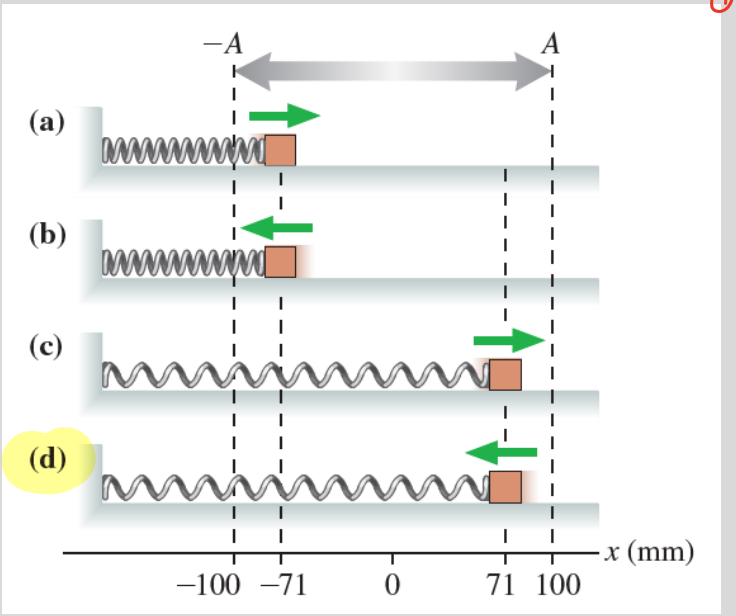
SHM and Circular Motion

Example: The figure below shows four oscillators at $t = 0$. Which one has the phase constant $\phi_0 = \frac{\pi}{4}\ rad$?

$x = A\cos(\phi_0 + \omega t)$

$v = -A\omega \sin(\phi_0 + \omega t)$

⊕ ⊖



SHM and Circular Motion

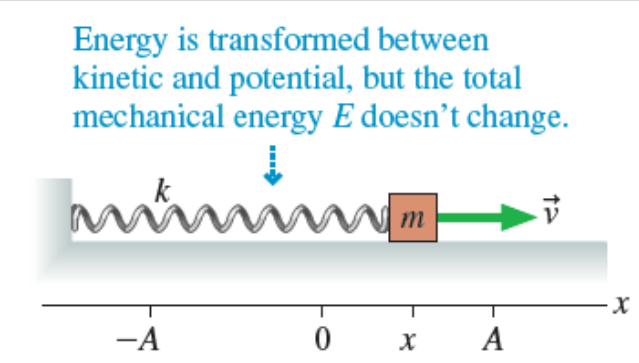
Example: An object on a spring oscillates with a period of 0.80s and an amplitude of 10 cm. At $t = 0$, it is 5.0 cm to the left of equilibrium and moving to the left. What are its position and velocity at $t = 2.0\ s$?

Energy in Simple Harmonic Motion

Relationship between the parameters of oscillations

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Energy in SHM



Conservation of energy

$$KE_i + PE_i = KE_f + PE_f$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

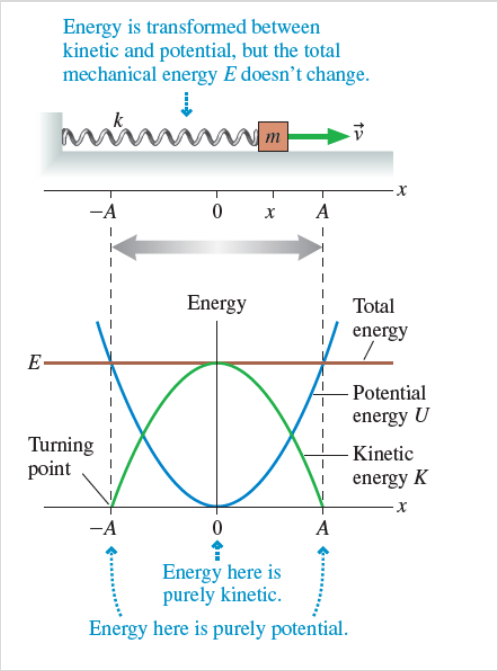
$$E(t) = KE + PE = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{const}$$

Energy in SHM

$$E(t) = KE + PE = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{const}$$

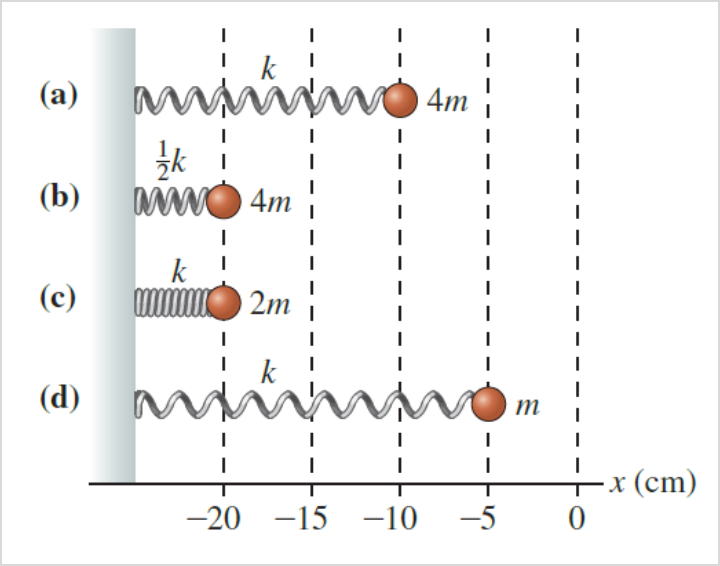
The graph of the **mechanical energy** of the mass

can be plotted as below



Energy in SHM

Example: The four springs below have been compressed from their equilibrium position at $x = 0\ cm$. When released, the attached mass will start to oscillate. Rank in order, from highest to lowest, the maximum speeds of the masses.



$$v = \sqrt{\frac{2PE}{m}}$$

a) $PE = \frac{1}{2}k\Delta x^2 = \frac{k}{2}(10)^2 = \frac{100}{2} = 50J$

$KE = 50J = \frac{1}{2}mv^2 \Rightarrow 100 = mv^2 \quad v = \sqrt{\frac{400}{4m}} = \sqrt{\frac{25}{m}} = \frac{5}{\sqrt{m}}$ K

b) $PE = \frac{1}{2}\left(\frac{1}{2}k\right)(20)^2 = \frac{1}{4}k\,400 = 100k\ J$

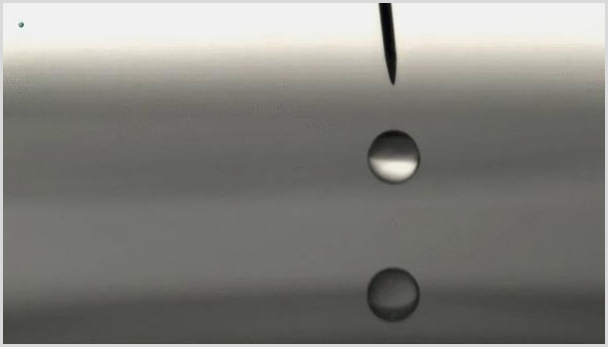
$v = \sqrt{\frac{2(100k)}{m}} = \frac{200\ K}{m}$

The Travelling Wave

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Travelling waves

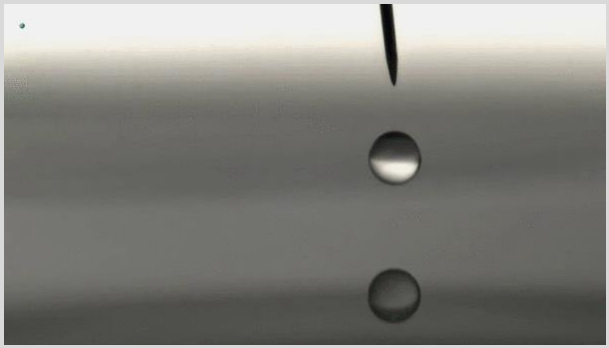
- Some examples of travelling wave in nature.

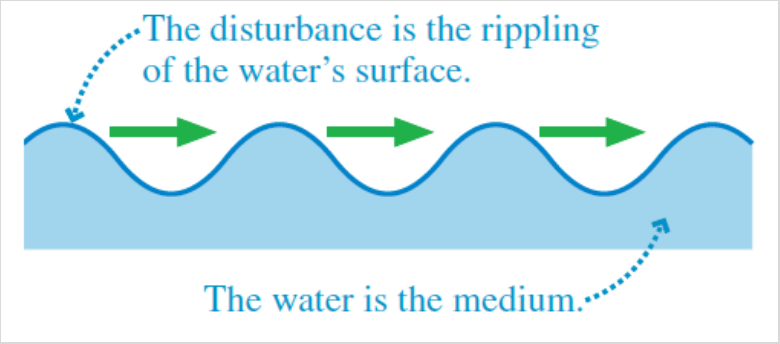




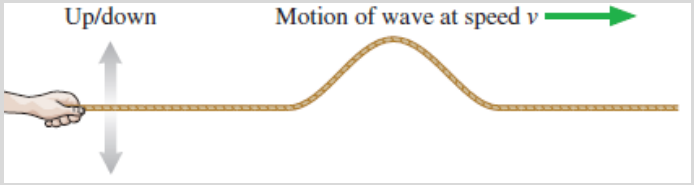
Travelling wave

- As a wave passes through a medium, the atoms of the medium are displaced from equilibrium, we call this a **disturbance** of the medium.
- A wave disturbance is created by **a source**. Once created, the disturbance travels outward through the medium at the **wave speed v**.
- A wave transfers energy, but it does not transfer any material or substance outward from the source.

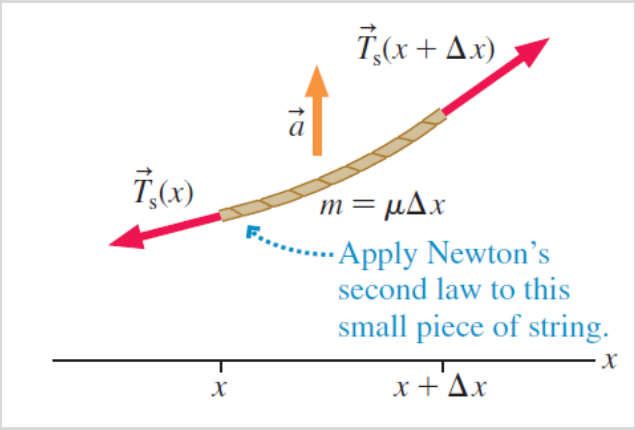




Travelling wave along a string



$$v_{string} = \sqrt{\frac{T_s}{\mu}}$$



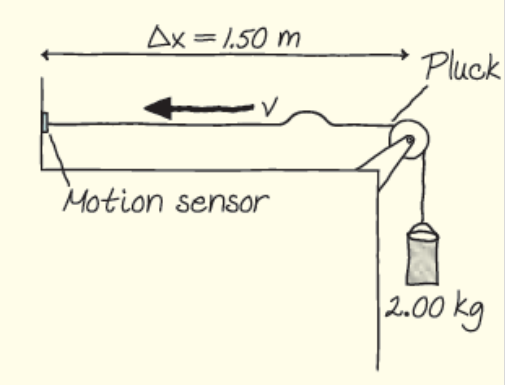
where:

T_s: tension force in the string

μ: mass per unit length (kg/m)

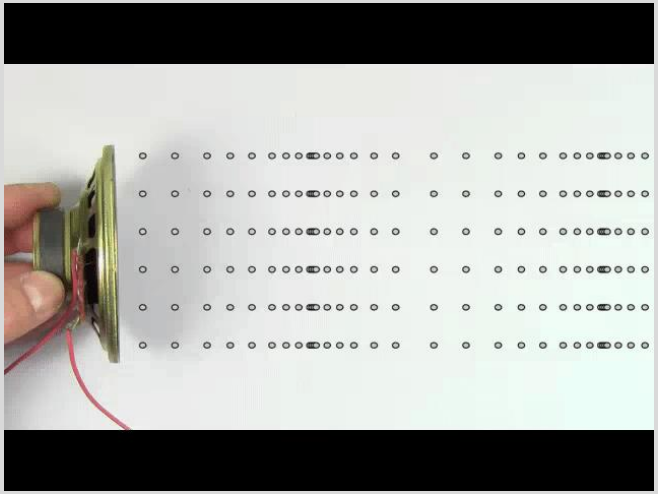
Travelling wave along a string

Example: A 2.0 meter long metal wire is attached to a motion sensor, stretched horizontally to a pulley 1.50 m away, then connected to a 2.0 kg hanging mass that provides tensions. A mechanical pick plucks a horizontal segment of wire right at the pulley, creating a small wave pulse that travels along the wire. What is the wire's linear density if the pulse takes 18.0 ms to travel the length of the wire?

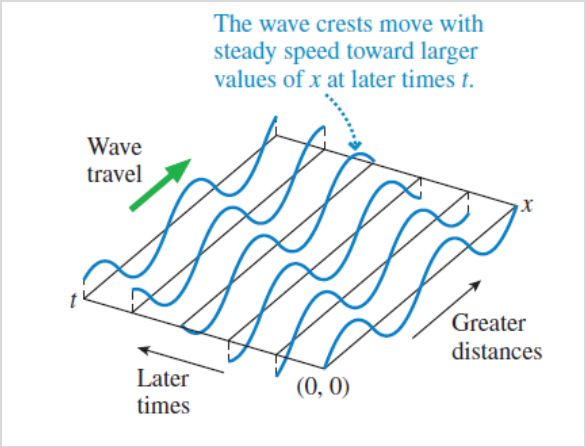


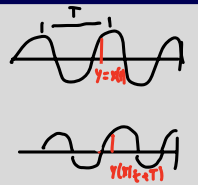
Sinusoidal Waves

- A wave source that oscillates with simple harmonic motion (SHM) generates a sinusoidal wave.
- This wave is a function of the **distance** x from the source and **time** t from the moment the source starts.



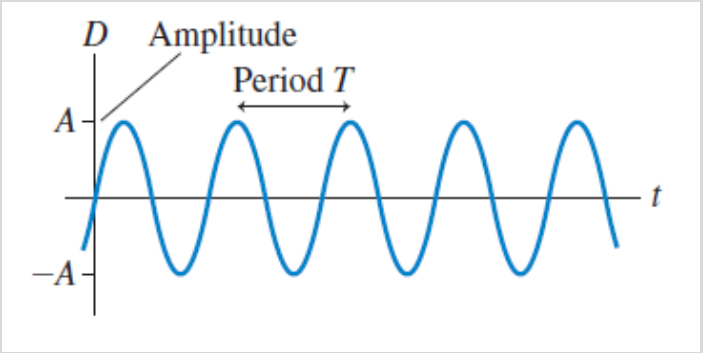
Sinusoidal Waves - Wavelength

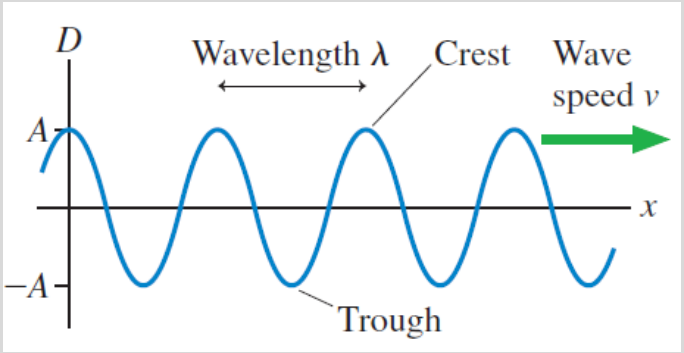




$y = y(x, t)$ Period & Wavelength

T λ





Sinusoidal Waves
Period and Wavelength

- There is an important relationship between the Period and Wavelength.
- During a time interval of exactly one period T, each crest of a sinusoidal wave travels forward a distance of exactly one wavelength λ.

$$v = \frac{\lambda}{T} = \lambda\frac{1}{T}$$

$$\rightarrow v = \lambda f$$

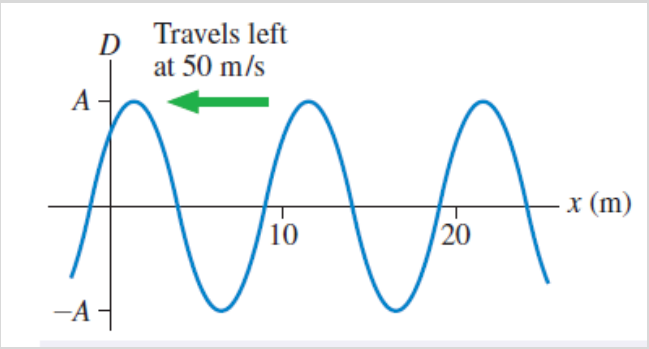
- However, it is more useful to write the equation above as:

$$\lambda = \frac{v}{f} = \frac{property\ of\ the\ medium}{property\ of\ the\ source}$$

The wavelength is a consequence of a wave of frequency f traveling through a medium in which the wave speed is v.

Sinusoidal Waves

Example: What is the frequency of this traveling wave?



a. 0.1 Hz
b. 0.2 Hz
c. 2 Hz
d. 5 Hz
e. 10 Hz
f. 500 Hz

$f = v/\lambda = \frac{50\,m/s}{10\,m} = 5\,Hz$

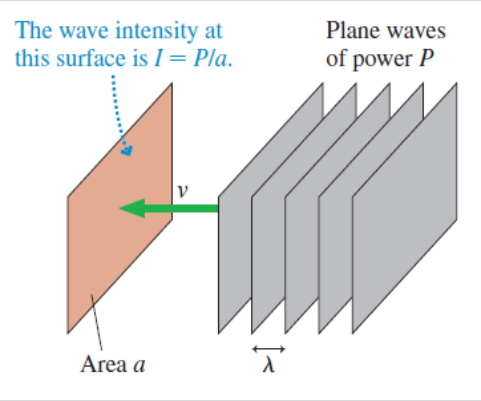
wave length similar to T
but measure space

Power, Intensity, and Decibels

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Power and Intensity

- Recall that a traveling wave transfers energy from one point to another.
- The power of a wave is the rate, in joules per second, at which the wave transfers energy.
- Consider the plane waves of power P impinge on area a, we define the **intensity of the wave** to be I which is given by:

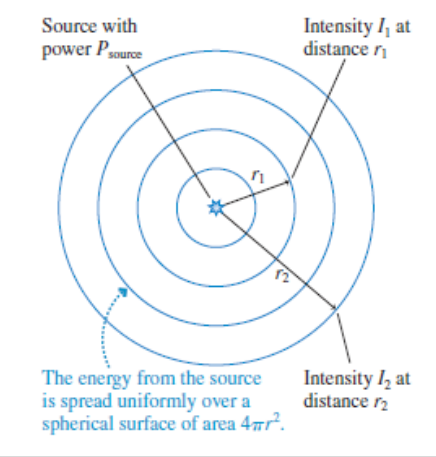


$$I = \frac{P}{a} \quad \left(\frac{W}{m^2}\right)$$

Power and Intensity

If the source of the wave radiates uniformly in all directions, the power at distance r is spread uniformly over the surface of a sphere of radius r. Hence, the Intensity of a uniform spherical wave is:

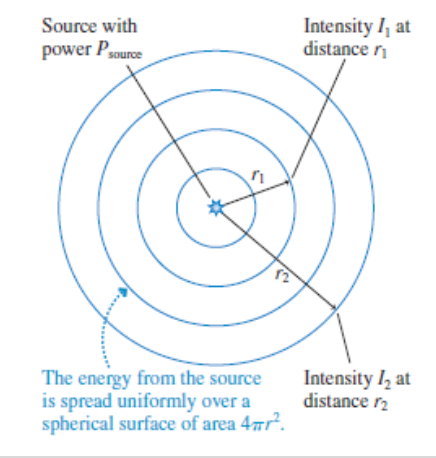
$$I = \frac{P}{4\pi r^2}$$



Power and Intensity

- Consider the intensity of the wave at 2 locations with radius r_1 with intensity $\mathrm{I}_1 = \frac{P}{4\pi r_1^2}$ and r_2 with intensity $\mathrm{I}_2 = \frac{P}{4\pi r_2^2}$, we deduce that:

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$$



Decibel

- For sound waves, human has 2 thresholds: threshold of hearing at $\mathbf{1} \times \mathbf{10^{-12}}\ \boldsymbol{W/m^2}$ and threshold of pain at $\mathbf{10}\ \boldsymbol{W/m^2}$**.**
- Hence, if we want to make a scale of loudness, it is convenient and logical to define a sound intensity level, expressed in **decibels (dB)**, as:

$$\beta = (10\ dB)\log_{10}\left(\frac{I}{I_0}\right)$$

Where $I_0 = 1.0 \times 10^{-12}\ W/m^2$

Power, Intensity, and Decibels

Example: The blender making a smoothie produces a sound intensity level of 83 dB. What is the intensity of the sound? What will the sound intensity level be if a second blender is turned on?